Arrays

1. Sum of Array

```java
package Arrays;
import java.util.*;

public class SumOfArray {
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        for (int i=0; i<n; i++){
            sum = sum + arr[i];
        }
        System.out.println(sum);
        sc.close();
    }
}
```

2. Mean of Array / Average of Array.

```
package Arrays;
import java.util.*;

public class MeanOfArray {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        for (int i=0; i<n; i++){
            sum = sum + arr[i];
        }
        System.out.println(sum / n);
        sc.close();
    }
}
```

3. Largest Number.

```
package Arrays;
import java.util.*;

public class LargestOfArray {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
        for (int i=0; i<n; i++){
            if (arr[i] > max){
                max = arr[i];
            }
        }
        System.out.println(max);
        sc.close();
    }
}
```

4. Second Largest.

```
package Arrays;
import java.util.*;
public class SecondLargest {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;
        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }
        for (int i=0; i<n; i++){
            if(arr[i] > largest){
                secondLargest = largest;
                largest = arr[i];
            }
            else if (arr[i] > secondLargest && arr[i] != largest){
                secondLargest = arr[i];
            }        }
        if (secondLargest == Integer.MIN_VALUE){
            System.out.println("No Second Largest");
        }
        else {
            System.out.println(secondLargest);
        }
        sc.close();
    }
}
```

5. Largest and Second Largest.

```java
package Arrays;
import java.util.Scanner;

public class LargestAndSecondLargest {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }

        for (int i=0; i<n; i++){
            if(arr[i] > largest){
                secondLargest = largest;
                largest = arr[i];
            }
            else if (arr[i] > secondLargest && arr[i] != largest){
                secondLargest = arr[i];
            }
        }

        if (secondLargest == Integer.MIN_VALUE){
            System.out.println("No Second Largest");
        }
        else {
            System.out.println(largest);
            System.out.println(secondLargest);
        }
        sc.close();
    }
}
```

6. Smallest Number.

```
package Arrays;
import java.util.*;

public class SmallestOfArray {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int min = arr[0];
        for (int i=0; i<n; i++){
            if (arr[i] < min){
                min = arr[i];
            }
        }
        System.out.println(min);
        sc.close();
    }
}
```

7. Second Smallest.

```
package Arrays;
import java.util.*;

public class SecondSmallest {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }

        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;

        for (int i=0; i<n; i++){
            if (arr[i] < smallest){
                secondSmallest = smallest;
                smallest = arr[i];
            }
            else if (arr[i] < secondSmallest && arr[i] != smallest){
                secondSmallest = arr[i];
            }
        }

        if (secondSmallest == Integer.MAX_VALUE){
                System.out.println("No Second Smallest");
            }
            else {
                System.out.println(secondSmallest);
            }
        sc.close();
    }
}
```

8. Smallest and Second Smallest.

```java
package Arrays;
import java.util.Scanner;

public class SmallestAndSecondSmallest {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }

        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;

        for (int i=0; i<n; i++){
            if (arr[i] < smallest){
                secondSmallest = smallest;
                smallest = arr[i];
            }
            else if (arr[i] < secondSmallest && arr[i] != smallest){
                secondSmallest = arr[i];
            }
        }

        if (secondSmallest == Integer.MAX_VALUE){
                System.out.println("No Second Smallest");
            }
            else {
                System.out.println(smallest);
                System.out.println(secondSmallest);
            }
        sc.close();
    }
}
```

9. Smallest and Largest.

```java
package Arrays;
import java.util.*;

public class SecondSmallestAndSecondLargest {
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;

        for (int i=0; i<n; i++){
            if (arr[i] > largest){
                secondLargest = largest;
                largest = arr[i];
            }
            else if (arr[i] > secondLargest && arr[i] != largest){
                secondLargest = arr[i];
            }
        }

        for (int i=0; i<n; i++){
            if (arr[i] < smallest){
                secondSmallest = smallest;
                smallest = arr[i];
            }
            else if(arr[i] < secondSmallest && arr[i] != smallest){
                secondSmallest = arr[i];
            }
        }

        if (secondLargest == Integer.MIN_VALUE){
            System.out.println("No Second Largest");
        }
```

```
        else {
            System.out.println(largest);
        }

        if (secondSmallest == Integer.MAX_VALUE){
            System.out.println("No Second Smallest");
        }
        else {
            System.out.println(smallest);
        }
        sc.close();
    }
}
```

10.Second Smallest and Second Largest.

```
package Arrays;
import java.util.*;

public class SecondSmallestAndSecondLargest {
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }

        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;

        for (int i=0; i<n; i++){
            if (arr[i] > largest){
                secondLargest = largest;
                largest = arr[i];
            }
            else if (arr[i] > secondLargest && arr[i] != largest){
                secondLargest = arr[i];
            }
```

```
        }

        for (int i=0; i<n; i++){
            if (arr[i] < smallest){
                secondSmallest = smallest;
                smallest = arr[i];
            }
            else if(arr[i] < secondSmallest && arr[i] != smallest){
                secondSmallest = arr[i];
            }
        }

        if (secondLargest == Integer.MIN_VALUE){
            System.out.println("No Second Largest");
        }
        else {
            System.out.println(secondLargest);
        }

        if (secondSmallest == Integer.MAX_VALUE){
            System.out.println("No Second Smallest");
        }
        else {
            System.out.println(secondSmallest);
        }
        sc.close();
    }
}
```

11. Smallest, Second Smallest, Largest, Second Largest.

```
package Arrays;
import java.util.*;

public class SecondSmallestAndSecondLargest {
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n < 2){
            System.out.println("Not Possible");
            // return;
        }
```

```
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;

        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;

        for (int i=0; i<n; i++){
            if (arr[i] > largest){
                secondLargest = largest;
                largest = arr[i];
            }
            else if (arr[i] > secondLargest && arr[i] != largest){
                secondLargest = arr[i];
            }
        }

        for (int i=0; i<n; i++){
            if (arr[i] < smallest){
                secondSmallest = smallest;
                smallest = arr[i];
            }
            else if(arr[i] < secondSmallest && arr[i] != smallest){
                secondSmallest = arr[i];
            }
        }

        if (secondLargest == Integer.MIN_VALUE){
            System.out.println("No Second Largest");
        }
        else {
             System.out.println(largest);
            System.out.println(secondLargest);
        }

        if (secondSmallest == Integer.MAX_VALUE){
            System.out.println("No Second Smallest");
        }
        else {
             System.out.println(smallest);
            System.out.println(secondSmallest);
        }
        sc.close();
    }
}
```

12. Remove Duplicates from Sorted Array.

```java
package Arrays;
import java.util.*;

public class RDSorted{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n <= 1){
            System.out.println(Arrays.toString(arr));
        }

        int idx = 1;
        for (int i=1; i<n; i++){
            if (arr[i] != arr[i-1]){
                arr[idx++] = arr[i];
            }
        }

        for (int i=0; i<idx; i++){
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```

13. Remove Duplicates from Unsorted Array.

```
package Arrays;
import java.util.*;

public class RDUnsorted {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        Set<Integer> set = new LinkedHashSet<>();

        for (int num : arr){
            set.add(num);
        }

        for (int num : set){
            System.out.print(num + " ");
        }
        sc.close();
    }
}
```

14. Reverse an Array.

```
import java.util.*;
class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();        }
        int i = 0;
        int j = arr.length - 1;
        while (i < j) {
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
            i++;
            j--;        }
        System.out.println(Arrays.toString(arr));
        sc.close();    }    }
```

15. Check if Array is Sorted.

```java
package Arrays;
import java.util.*;

public class IsSorted {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        boolean isSorted = true;

        for(int i=1; i<n; i++){
            if (arr[i] < arr[i-1]){
                isSorted = false;
            }
        }

        if (isSorted){
            System.out.println("Sorted");
        }
        else {
            System.out.println("Not Sorted");
        }
        sc.close();
    }
}
```

16. Maximum Consecutive Bit.

```java
package Arrays;
import java.util.*;

public class MaxBit {
    public static void main (String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n == 0){
            System.out.println(0);
        }

        int count =1;
        int ans = 0;

        for (int i=1; i<n; i++){
            if (arr[i] == arr[i-1]){
                count++;
            }
            else {
                ans = Math.max(ans, count);
                count = 1;
            }
        }
        System.out.println(Math.max(ans, count));
        sc.close();
    }
}
```

17. Subset of an Array.

```
package Arrays;
import java.util.*;

public class ArraySubset{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int m = sc.nextInt();
        int[] a = new int[m];
        for (int i=0; i<m; i++){
            a[i] = sc.nextInt();
        }

        int n = sc.nextInt();
        int[] b = new int[n];
        for (int i=0; i<n; i++){
            b[i] = sc.nextInt();
        }

        Arrays.sort(a);
        Arrays.sort(b);

        int i = 0;
        int j = 0;

        while(i<m && j<n){
            if (a[i] < b[j]){
                i++;
            }
            else if (a[i] == b[j]){
                i++;
                j++;
            }
            else{
                System.out.println(false);
                return;
            }
            sc.close();
        }
        System.out.println(j == n);
    }
}
```

18. Rotate an Array.

```
package Arrays;
import java.util.*;

public class RotateArray{
    static void reverse (int[] arr, int i, int j){
        while (i < j){
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
            i++;
            j--;
        }

    }

    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n= sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int d = sc.nextInt();

        if (n == 0){
            System.out.println(0);
            // return;
        }

        d = d % n;

        reverse(arr,0, d-1);
        reverse(arr, d, n-1);
        reverse(arr, 0, n-1);

        for (int i=0; i<n; i++){
            System.out.print(arr[i]+ " ");
        }
        sc.close();
    }
}
```

19. Equilibrium.

```
package Arrays;
import java.util.*;

public class Equilibrium{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        for(int num : arr){
            sum += num;
        }

        int left = 0;
        boolean found = false;

        for(int i=0; i<n; i++){
            sum -= arr[i];
            if (left == sum){
                found = true;
                break;
            }
            left += arr[i];
        }
        System.out.println(found);
        sc.close();
    }
}
```

20. Buy and Sell the Stock.

```java
package Arrays;
import java.util.*;

public class BuySell {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] prices = new int[n];
        for (int i=0; i<n; i++){
            prices[i] = sc.nextInt();
        }

        if (n == 0){
            System.out.println(0);
            // return;
        }

        int mini = 1;
        int maxp = 0;

        for(int i=0; i<n; i++){
            if(prices[i] < mini){
                mini = prices[i];
            }

            int profit = prices[i] - mini;
            if (profit>maxp){
                maxp = profit;
            }
        }
        System.out.println(maxp);
        sc.close();
    }
}
```

21. Matrix Addition.

```
package Arrays;
import java.util.*;

public class MatrixAdd {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int m = sc.nextInt();

        int a[][] = new int[n][m];
        int b[][] = new int[n][m];

        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                a[i][j] = sc.nextInt();
            }
        }

        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                b[i][j] = sc.nextInt();
            }
        }

        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                a[i][j] += b[i][j];
            }
        }

        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                System.out.print(a[i][j] + " ");
            }
        }
        sc.close();
    }
}
```

22. Union of 2 Arrays.

```
package Arrays;
import java.util.*;

public class Union {
    public static void main(String[] args){
        Scanner sc =  new Scanner(System.in);

        int m = sc.nextInt();
        int[] a = new int[m];
        for(int i=0; i<m; i++){
            a[i] = sc.nextInt();
        }

        int n = sc.nextInt();
        int[] b = new int[n];
        for(int i=0; i<n; i++){
            b[i] = sc.nextInt();
        }

        Set<Integer> set = new LinkedHashSet<>();

        for(int num : a){
            set.add(num);
        }

        for(int num : b){
            set.add(num);
        }

        for(int num : set){
            System.out.print(num+ " ");
        }
        sc.close();
    }
}
```

23. Median of an Array.

```
package Arrays;
import java.util.*;

public class Median{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        if (n == 0){
            System.out.println("No Elements");
        }

        int mid = n / 2 - 1;
        int mid1 = n / 2;

        if (n%2 == 0){
            System.out.println((arr[mid] + arr[mid1]) / 2);
        }
        else {
            System.out.println(arr[n/2]);
        }
        sc.close();
    }
}
```

24. Count Sorted Rows.

```
package Arrays2;
import java.util.*;

public class CountSortedRows{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        int a[][] = new int[n][m];
        for(int i=0; i<n; i++){
            for (int j=0; j<m; j++){
                a[i][j] = sc.nextInt();
            }
        }
        int count = 0;
        for(int i=0; i<n; i++){
            boolean inc = true;
            boolean dec = true;

            for (int j=0; j<m-1; j++){
                if(a[i][j] <= a[i][j+1]){
                    dec = false;
                }
                if(a[i][j] >= a[i][j+1]){
                    inc = false;
                }
            }
            if(inc || dec){
                count++;
            }
        }
        System.out.println(count);
        sc.close();
    }
}
```

25. Rows with Maximum 1's.

```
package Arrays2;
import java.util.*;

public class Maximum1{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        int a[][] = new int[m][n];
        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                a[i][j] = sc.nextInt();
            }
        }
        int i = 0;
        int j = m - 1;
        int k = -1;
        while (i<n && j>=0){
            if(a[i][j] == 1){
                k = i;
                j--;
            }
            else{
                j++;
            }
        }
        System.out.println(k);
        sc.close();
    }
}
```

26. Frequencies in Limited Array.

```java
package Arrays2;
import java.util.*;

public class Maximum1{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        int a[][] = new int[m][n];
        for(int i=0; i<n; i++){
            for(int j=0; j<m; j++){
                a[i][j] = sc.nextInt();
            }
        }
        int i = 0;
        int j = m - 1;
        int k = -1;
        while (i<n && j>=0){
            if(a[i][j] == 1){
                k = i;
                j--;
            }
            else{
                j++;
            }
        }
        System.out.println(k);
        sc.close();
    }
}
```

27. Non Repeating Elements.

```java
package Arrays2;
import java.util.*;

public class NonRepeatEle{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        Map<Integer, Integer> map = new HashMap<>();

        for (int x : arr){
            map.put(x, map.getOrDefault(x, 0) + 1);
        }

        for (int key : map.keySet()){
            if(map.get(key) == 1){
                System.out.print(key + " ");
            }
        }
        sc.close();
    }
}
```

28. Repeating Elements.

```java
package Arrays2;
import java.util.*;

public class RepeatingEle{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        Map<Integer, Integer> map = new HashMap<>();

        for (int x : arr){
            map.put(x, map.getOrDefault(x, 0) + 1);
        }

        for (int key : map.keySet()){
            if(map.get(key) > 1){
                System.out.print(key + " ");
            }
        }
        sc.close();
    }
}
```

29. Replace by Rank.

```java
package Arrays2;
import java.util.*;

public class ReplaceByRank{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int[] sorted = arr.clone();
        Arrays.sort(sorted);

        Map<Integer, Integer> map = new HashMap<>();
        int rank = 1;

        for (int i=0; i<n; i++){
            if (i>0 && sorted[i] == sorted[i-1]){
                continue;
            }
            map.put(sorted[i], rank++);
        }

        for (int i=0; i<n; i++){
            System.out.print(map.get(arr[i]) + " ");
        }
        sc.close();
    }
}
```

30. Kadane's Max Sum Subarray.

```
package Arrays2;
import java.util.*;

public class KadaneSum{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int sum = 0;
        int ans = Integer.MIN_VALUE;

        for (int i=0; i<n; i++){
            sum = Math.max(arr[i], sum + arr[i]);
            ans = Math.max(ans, sum);
        }
        System.out.println(ans);
        sc.close();
    }
}
```

31. Kadane's Max Product Subarray.

```java
package Arrays2;
import java.util.*;

public class KadaneProduct{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
        int min = arr[0];
        int ans = arr[0];

        for (int i=1; i<n; i++){
            if (arr[i] < 0){
                int temp = max;
                max = min;
                min = temp;
            }

            max = Math.max(arr[i], max * arr[i]);
            min = Math.max(arr[i], min * arr[i]);

            ans = Math.max(ans, max);
        }
        System.out.println(ans);
        sc.close();
    }
}
```

32. Sort According to Array.

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        Integer[] arr1 = {2,1,2,5,7,1,9,3,6,8,8};
        int[] arr2 = {2,1,8,3};

        Map<Integer, Integer> order = new HashMap<>();

        for (int i = 0; i < arr2.length; i++) {
            order.put(arr2[i], i);
        }

        Arrays.sort(arr1, (a, b) -> {
            boolean aPresent = order.containsKey(a);
            boolean bPresent = order.containsKey(b);

            if (aPresent && bPresent)
                return order.get(a) - order.get(b);

            if (aPresent)
                return -1;

            if (bPresent)
                return 1;

            return a - b;
        });

        for (int x : arr1) {
            System.out.print(x + " ");
        }
    }
}
```

33. Unique Elements.

```
package Arrays2;
import java.util.*;

public class UniqueElements{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int k = sc.nextInt();

        Map<Integer, Integer> freq = new HashMap<>();
        ArrayList<Integer> res = new ArrayList<>();

        for (int i=0; i<k; i++){
            freq.put(arr[i], freq.getOrDefault(arr[i], 0) + 1);
        }
        res.add(freq.size());

        for (int i=k; i<n; i++){
            freq.put(arr[i], freq.getOrDefault(arr[i], 0) + 1);
            freq.put(arr[i-k], freq.get(arr[i-k]) - 1);
            if(freq.get(arr[i-k]) == 0){
                freq.remove(arr[i-k]);
            }
            res.add(freq.size());
        }

        for (int x : res){
            System.out.print(x + " ");
        }
        sc.close();
    }
}
```

34. First Negative in the Window.

```java
package Arrays2;
import java.util.*;

class FirstNegInK {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int k = sc.nextInt();

        ArrayList<Integer> res = new ArrayList<>();
        int neg = 0;

        for (int i=k-1; i<n; i++){
            while (neg < n && (neg < i-k+1 || arr[neg] >= 0)){
                neg++;
            }

            if (neg <= i && arr[neg] < 0){
                res.add(arr[neg]);
            }
            else {
                res.add(0);
            }
        }

        for (int x : res){
            System.out.print(x + " ");
        }
        sc.close();
    }
}
```

35. Trapping Rain Water.

```
package Arrays2;
import java.util.*;

public class RainWaterTrap {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int left = 0;
        int leftMax = 0;
        int right = n - 1;
        int rightMax = 0;
        int res = 0;

        while (left <= right){
            if (arr[left] <= arr[right]){
                if (arr[left] >= leftMax){
                    leftMax = arr[left];
                }
                else {
                    res = res + (leftMax - arr[left]);
                }
                left++;
            }
            else {
                if (arr[right] >= rightMax){
                    rightMax = arr[right];
                }
                else {
                    res = res + (rightMax - arr[right]);
                }
                right--;
            }
        }
        System.out.println(res);
        sc.close();
    }
}
```

36. Maximum Sum Subrray of size K.

```java
package Arrays2;
import java.util.*;

public class MaxSubarrayOfK{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int k = sc.nextInt();

        int sum = 0;
        int max = Integer.MIN_VALUE;

        for (int i=0; i<n; i++){
            sum = sum + arr[i];

            if (i >= k){
                sum = sum - arr[i-k];
            }

            if (i >= k -1){
                max = Math.max(max, sum);
            }
        }
        System.out.println(max);
        sc.close();
    }
}
```

37. Maximum Average Subarray.

```java
package Arrays2;
import java.util.*;

public class MaxAvgSubarray{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }
        int k = sc.nextInt();

        int sum = 0;
        for (int i=0; i<k; i++){
            sum += arr[i];
        }

        int max = sum;
        int index = 0;
        for (int i=k; i<n; i++){
            sum += arr[i] - arr[i-k];
            if (sum > max){
                max = sum;
                index = i - k + 1;
            }
        }
        System.out.println(index);
        sc.close();
    }
}
```

38. Most Water Container.

```
import java.util.*;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] height = new int[n];

        for (int i = 0; i < n; i++) {
            height[i] = sc.nextInt();
        }

        int left = 0;
        int right = n - 1;

        int maxArea = 0;

        while (left < right) {

            int width = right - left;
            int h = Math.min(height[left], height[right]);

            maxArea = Math.max(maxArea, width * h);

            if (height[left] < height[right]) {
                left++;
            } else {
                right--;
            }
        }

        System.out.println(maxArea);

        sc.close();
    }
}
```

39. Sort in IO-DO.

```java
package Arrays2;
import java.util.*;

public class SortIODO {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr=  new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        Arrays.sort(arr);

        int mid = (n + 1) / 2;
        int left = mid;
        int right = n - 1;

        while (left < right){
            int temp = arr[left];
            arr[left] = arr[right];
            arr[right] = temp;
            left++;
            right--;
        }

        for (int x : arr){
            System.out.print(x + " ");
        }
        sc.close();
    }
}
```

40. Add Element to an Array.

```java
package Arrays2;
import java.util.*;

public class AddEle {
    public static void main (String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        ArrayList<Integer> list = new ArrayList<>();
        for (int i=0; i<n; i++){
            list.add(sc.nextInt());
        }

        int ele = sc.nextInt();
        int pos = sc.nextInt();

        list.add(pos, ele);

        for (int x : list){
            System.out.print(x + " ");
        }
        sc.close();
    }
}
```

41.Symmetric Pairs.

```java
package Arrays2;
import java.util.*;

public class SymPairs {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[][] pairs = new int[n][2];
        for(int i=0; i<n; i++){
            pairs[i][0] = sc.nextInt();
            pairs[i][1] = sc.nextInt();
        }

        Map<Integer, Integer> map = new HashMap<>();

        for(int i=0; i<n; i++){
            int a = pairs[i][0];
            int b = pairs[i][1];

            if(map.containsKey(b) && map.get(b) == a){
                System.out.print(a + " ");
            }
            else{
                map.put(a, b);
            }
        }
        sc.close();
    }
}
```

42. Search an Element in Array.

```
package Arrays2;
import java.util.*;

public class SearchEle {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int[] arr = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int ele = sc.nextInt();

        int index = -1;
        for(int i=0; i<n; i++){
            if(arr[i] == ele){
                index = i;
                break;
            }
        }
        System.out.println(index);
        sc.close();
    }
}
```

43. Intersection of 2 Arrays.

```
package Arrays;
import java.util.*;

public class Intersection {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int m = sc.nextInt();
        int[] a = new int[m];
        for (int i=0; i<m; i++){
            a[i] = sc.nextInt();
        }

        int n = sc.nextInt();
        int[] b = new int[n];
        for (int i=0; i<n; i++){
            b[i] = sc.nextInt();
        }

        Arrays.sort(a);
        Arrays.sort(b);

        int i = 0;
        int j = 0;

        while (i<m && j<n){
            if(a[i] < b[j]){
                i++;
            }
            else if (a[i] > b[j]){
                j++;
            }
            else{
                System.out.print(a[i] + " ");
                i++;
                j++;
            }
        }
        sc.close();
    }
}
```

44. Move Zeros to End.

```
package Arrays;
import java.util.*;

public class Move0toEnd {
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int arr[] = new int[n];
        for (int i=0; i<n; i++){
            arr[i] = sc.nextInt();
        }

        int idx = 0;
        for(int i=0; i<n; i++){
            if (arr[i] != 0){
                arr[idx++] = arr[i];
            }
        }

        while (idx<n){
            arr[idx++] = 0;
        }

        for (int i=0; i<n; i++){
            System.out.print(arr[i] + " ");
        }
        sc.close();
    }
}
```